

Mahsa Ghasemi

Ph.D. Candidate in Computer Science | Indiana University Bloomington

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RESEARCH PROFILE

Computer vision and biomedical AI researcher specializing in computational pathology, medical image analysis, and deep learning. Current work focuses on nucleus and cell detection/classification, morphology-aware supervision, histopathology and microscopy image analysis, and representation learning with limited annotations. Experienced with Ki-67 pathology, spatial transcriptomics, multimodal biomedical data, and rigorous evaluation of detection, segmentation, and classification systems.

RESEARCH INTERESTS

Medical Image Analysis • Computer Vision • Computational Pathology • Deep Learning • Image Segmentation • Cell/Nucleus Detection and Classification • Self-Supervised Learning • Morphology-Aware Learning

EDUCATION

Doctor of Philosophy (Ph.D.), Computer Science | *Indiana University Bloomington, USA* | 2023-2028 (expected)

Master of Science (M.S.), Computer Engineering | *Shahrood University of Technology, Iran* | 2018-2021

Bachelor of Science (B.S.), Computer Engineering | *Quchan University of Technology, Iran* | 2014-2018

RESEARCH & TEACHING EXPERIENCE

Associate Instructor, Indiana University | 2024-Present

- Conduct research in computational methods for healthcare, with emphasis on medical imaging, machine learning, gene-expression analysis, and spatial imaging.
- Support student learning and mentor students in data analysis and computational problem solving.

Research Assistant, Indiana University | 2023-2024

- Worked on biomedical data analysis, image processing, machine learning, spatial gene expression, single-cell sequencing, and bioinformatics-integrated imaging.

SELECTED RESEARCH PROJECTS

ShadoNet - Cell Detection and Classification for Ki-67 Pathology Images | 2025-Present

- Developed a morphology-aware deep learning framework for cell/nucleus detection and classification in Ki-67-stained histopathology images.
- Combined pixel-level proximity representations with object-level shape guidance and geometric supervision, including scaled IoU and Hausdorff-distance-based objectives.
- Used SAM-derived masks and centroid-based filtering to construct shape-aware training targets; evaluated against modern nucleus-analysis baselines.
- Manuscript accepted for publication in Bioinformatics (Oxford University Press).

Label-Free Sporulation Stage Detection in Phase-Contrast Microscopy | Current

- Developing a center-based computer vision framework to detect and classify *Bacillus subtilis* cells as vegetative, developing, or mature spores from phase-contrast microscopy.
- Designed GT-anchored evaluation for detection and classification, emphasizing per-class and macro F1 under class imbalance.

Morphology-Guided Foundation Models for Neuroendocrine Neoplasms | Current

- Investigating self-supervised histopathology representation learning for differentiating high-grade neuroendocrine neoplasms using H&E and Ki-67 images with limited labeled data.

Spatial Gene Expression Alignment via Optimal Transport | 2024

- Applied graph-based optimal transport to align spatial transcriptomics datasets across developmental stages and integrated gene-expression and spatial features for cellular trajectory and morphology analysis.

PUBLICATIONS

- Ghasemi, M., Zhang, X. “ShadoNet: A Cell Detection and Classification Framework for Ki-67 Pathology Images.” Bioinformatics, accepted for publication, 2026.
- Ghasemi, M., Hassanpour, H. “FRIH: A Face Recognition Framework Using Image Hashing.” Multimedia Tools and Applications, 2024. DOI: 10.1007/s11042-023-18007-9.
- Ghasemi, M., Hassanpour, H. “A Three-stage Filtering Approach for Face Recognition.” International Journal of Engineering, 34(8), 1856-1864, 2021. DOI: 10.5829/IJE.2021.34.08B.06.
- Ghasemi, M., Jafarinejad, F. “A Persian Text Classification Model Based on a Combination of Gaussian and Bayesian Approaches.” 5th Iranian Conference on Signal Processing and Intelligent Systems, 2019.
- Mortezaie, Z., Ghasemi, M., Fateh, M. “Predicting Room Temperature Using Reinforcement Learning.” 5th Iranian Conference on Signal Processing and Intelligent Systems, 2019.

PEER REVIEW

Peer Reviewer - International Journal of Engineering | 2022
Reviewed a manuscript on comparative analysis of K-Means and K-Medoids for customer segmentation using the RFM model.

TECHNICAL SKILLS

Programming	Python, MATLAB
Deep Learning	PyTorch, TensorFlow, MONAI; CNNs, U-Net-style models, Vision Transformers
Computer Vision	OpenCV; segmentation, detection, classification, morphology-aware learning
Biomedical AI	Histopathology, Ki-67 imaging, microscopy, spatial transcriptomics, single-cell imaging
Data & Computing	NumPy, Pandas, Matplotlib, Git, Jupyter, HPC workflows
Methods	Multi-task learning, self-supervised learning, optimal transport, graph-based modeling, geometric losses
Evaluation	Precision, Recall, F1, Dice, AJI; cross-validation and reproducible ML evaluation